

Drop Down List and its Methods



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Introduction

- In Selenium, the Select class provides the implementation of the HTML SELECT tag.
- A Select tag provides the helper methods with select and deselect options.
- The Select is a Webdriver class that can work with the HTML SELECT tag.
- It exposes several “Select By” and “Deselect By” type methods. You can apply these methods to change the value in the dropdown and multiple select fields
- Import Select class

from selenium.webdriver.support.ui import Select

Introduction

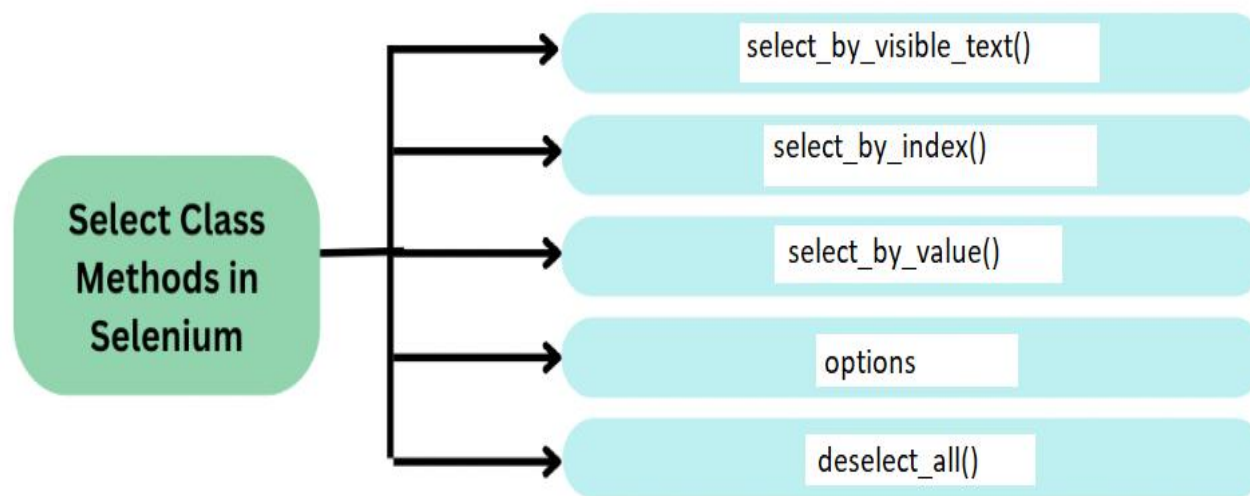
- First locate a <select> element, then use it to initialize a Select object.
- Example:

```
select_element = driver.find_element(By.Name, value="selproduct")  
select = Select(select_element)
```

Different Select Class Methods

Different Select Class Methods

- The following are the most commonly used select class methods in selenium to deal with a drop-down list:



Different Select Class Methods

1. select_by_visible_text()

- This select class method is used to select one of the options in a drop-down box or an option among multiple selection boxes.
- It takes a parameter of String which is one of the values of Select element and it returns nothing.

Example:

```
select_element = driver.find_element(By.ID, value="drop1")
```

```
select = Select(select_element)
```

```
select.select_by_visible_text("doc 2")
```

Different Select Class Methods

1. select_by_visible_text()

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="UTF-8">
    <title>Select Example By Visible Text</title>
  </head>
  <body>
    <html>
      <body>
        <select name="Browsers">
          <option value="0" selected>Select...</option>
          <option value="1">Chrome</option>
          <option value="2">FireFox</option>
          <option value="3">IE</option>
          <option value="4">Safari</option>
          <option value="5">Opera</option>
        </select>
      </body>
    </html>
  </body>
</html>
```

Different Select Class Methods

1. `select_by_visible_text()`

```
select_element = driver.find_element(By.NAME, value="Browsers")
```

```
select = Select(select_element)
```

```
select.select_by_visible_text("Chrome")
```


Different Select Class Methods

2. select_by_index()

- This method is similar to 'select_by_visible_text()', but the difference here is that the user has to provide the index number for the option rather than text.
- It takes the integer parameter which is the index value of Select element and it returns nothing.

Example:

```
select_element = driver.find_element(By.ID, value="drop1")
```

```
select = Select(select_element)
```

```
select.select_by_index(1)
```

Different Select Class Methods

1. `select_by_index()`

```
select_element = driver.find_element(By.NAME, value="Browsers")
```

```
select = Select(select_element)
```

```
select.select_by_index(2)
```

Different Select Class Methods

3. select_by_value()

- This method asks for the value of the desired option rather than the option text or an index. It takes a String parameter which is one of the values of Select element and it does not return anything.

Example:

```
select_element = driver.find_element(By.ID, value="drop1")
```

```
select = Select(select_element)
```

```
select.select_by_value("apple")
```

Different Select Class Methods

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="UTF-8">
    <title>Select Example By Value</title>
  </head>
  <body>
    <html>
      <body>
        <select name="Browsers">
          <option value="0" selected>Select...</option>
          <option value="Chrome">Chrome</option>
          <option value="FireFox">FireFox</option>
          <option value="IE">IE</option>
          <option value="Safari">Safari</option>
          <option value="Opera">Opera</option>
        </select>
      </body>
    </html>
  </body>
</html>
```

Different Select Class Methods

3. select_by_value()

```
select_element = driver.find_element(By.NAME, value="Browsers")
```

```
select = Select(select_element)
```

```
select.select_by_value("opera")
```

Different Select Class Methods

4. options()

- This method gets all the options belonging to the Select tag. It takes no parameter and returns List<WebElements>.

Example:

```
select_element = driver.find_element(By.ID, value="drop1")
```

```
select = Select(select_element)
```

```
dropdown_options = select.options
```

Different deselect Methods

deselect Methods

1. `deselect_by_index` Method : `select.deselect_by_index(3)`

- Use it when you want to deselect any option via its index. It accepts the index as an argument.

2. `deselect_by_value` Method: `select.deselect_by_value("Opera")`

- Use it when you wish to deselect all options using a value that matches the value in the dropdown options.

3. `deselect_by_visible_text` Method : `select.deselect_by_visible_text("selenium")`

- Use it to deselect all options by supplying the input text that matches the text in the dropdown options.

Different Select Class Methods

deselect Methods

4. deselect_all Method: `select.deselect_all()`

- Usage: Use it to turn off all the selected entries at once. But you can use this method if the `<Select>` tag has `<multiple>` attributes enabled in the HTML.
- Otherwise, it'll throw the `<NotImplemented>` exception. Hence, it's compulsory to have an attribute like `<multiple="multiple">` while defining the `<Select>` tag.

Drop Down List and Methods

An Example Scenario – single select option

```
import time
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import Select

driver = webdriver.Chrome()
driver.maximize_window()
driver.get("https://omayo.blogspot.com/")
select_element = driver.find_element(By.ID, value="drop1")
select = Select(select_element)

dropdown_options = select.options

print(len(dropdown_options))
for option in dropdown_options:
    print(option.text)

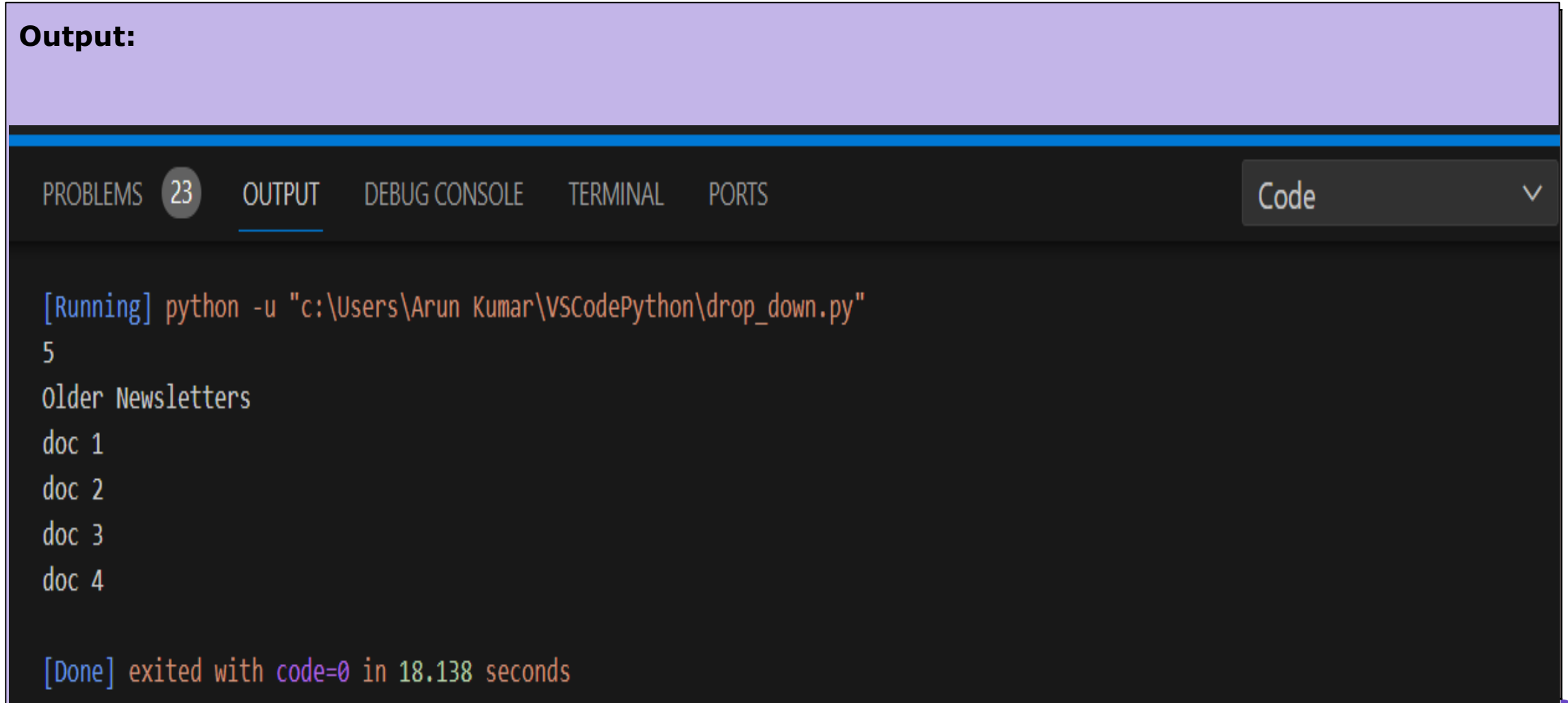
select.select_by_visible_text("doc 2")
#select.select_by_index(4)
#select.select_by_value("mno")

time.sleep(5)
```

Why do we need selenium waits?

An Example Scenario

Output:



The screenshot shows the VS Code interface with the 'OUTPUT' tab selected. The output text is as follows:

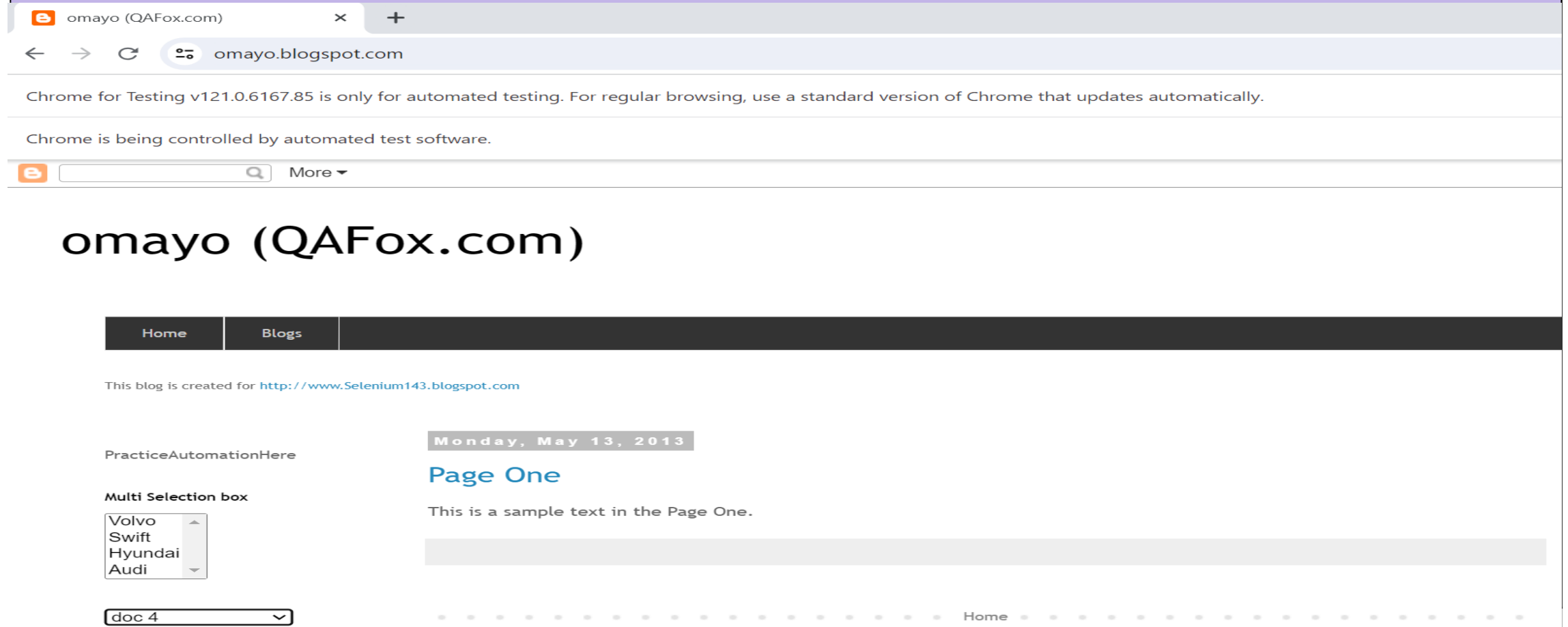
```
[Running] python -u "c:\Users\Arun Kumar\VSCodePython\drop_down.py"
5
Older Newsletters
doc 1
doc 2
doc 3
doc 4

[Done] exited with code=0 in 18.138 seconds
```

Why do we need selenium waits?

An Example Scenario

Output:



Drop Down List and Methods

An Example Scenario – Multi-select option

```
import time
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import Select

driver = webdriver.Chrome()
driver.maximize_window()
driver.get("https://omayo.blogspot.com/")
select_element = driver.find_element(By.ID, value="multiselect1")

select = Select(select_element)
multiselect_options = select.options

print(len(multiselect_options))
for option in multiselect_options:
    print(option.text)

select.select_by_visible_text("Swift")
select.select_by_index(0)
select.select_by_value("audix")

time.sleep(5)
```

Drop Down List and Methods

An Example Scenario – Multi-select option

```
select.deselect_by_visible_text("Swift")
select.deselect_by_index(0)
select.deselect_by_value("audix")
#select.deselect_all()
time.sleep(5)

select.select_by_visible_text("Hyundai")
```

Why do we need selenium waits?

An Example Scenario

Output:

PROBLEMS

22

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Code

```
[Running] python -u "c:\Users\Arun Kumar\VSCodePython\drop_down.py"
```

```
4
```

```
Volvo
```

```
Swift
```

```
Hyundai
```

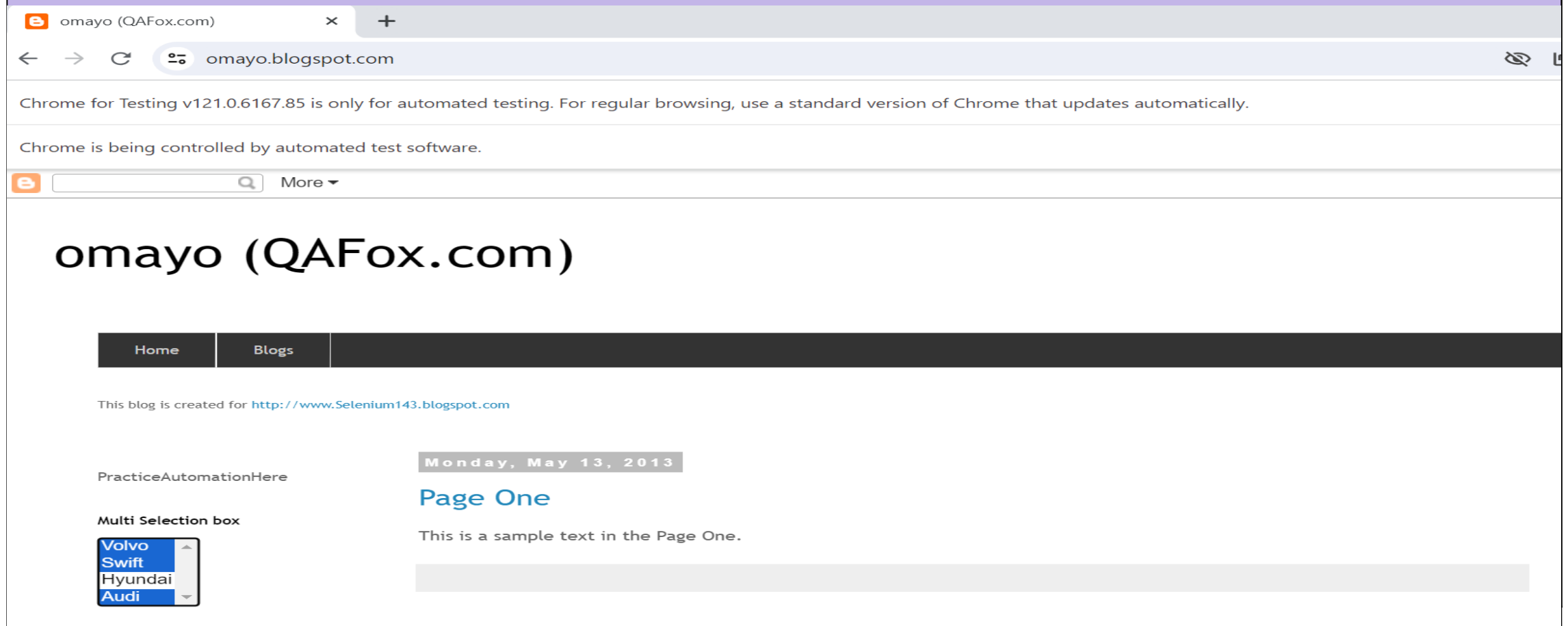
```
Audi
```

```
[Done] exited with code=0 in 21.61 seconds
```

Why do we need selenium waits?

An Example Scenario

Output:



How to Select Value & Handle Dropdown in Selenium without Select Class



Why do we need selenium waits?

Introduction

- Drop downs in a website could be created in several different ways.
- Some dropdowns are created using `<select>` HTML tag and some others are created using `<ul>` ,`<li>`, `<button>` and `<div>` tags.
- Some dropdowns are dynamic in nature which means after clicking or selecting any option, the drop downs values would populate accordingly.
- And in some cases, you need to mouse hover on an element in Selenium to see the drop-down options.
-

Why do we need selenium waits?

Introduction

- Selenium WebDriver provides Select class which can be used only for drop down created using <select> HTML tag.
- Select class has methods such as select_by_visible_text(), select_by_index() and select_by_value() to select the desired option from dropdown.
- However, for NonSelect dropdowns, Select class cannot be used.

Why do we need selenium waits?

How to Select Dropdown in Selenium?

You can handle dropdown in Selenium using Select Class and without using Select Class. Below are 5 different methods that can be used to select value in dropdown in Selenium without using Select Class. These methods are:

- 1.By storing all the options in List and iterating through it
- 2.By creating Custom Locator and without iterating the List
- 3.By using JavaScriptExecutor class
- 4.By using sendKeys
- 5.By using ActionsChains Class

Why do we need selenium waits?

How to Select Dropdown in Selenium?

1.By storing all the options in List and iterating through it

- You can handle Dropdown in Selenium by storing all the options in the form of a List and then iterate through it

Thread.Sleep

An Example Scenario

```
import time
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver import ActionChains

driver = webdriver.Chrome()
driver.maximize_window()

driver.get("https://www.bstackdemo.com/")
driver.find_element(By.XPATH, value="//select").click()

#Method 1: By storing all the options in List and iterating through it

allOptions = driver.find_elements(By.CSS_SELECTOR, value="select option")
print("Options Size:", +len(allOptions))
option = "Lowest to highest"
for options in allOptions:
    if options.text.__eq__(option):
        options.click()
        print("clicked")
        break

time.sleep(5)
driver.quit()
```

Why do we need selenium waits?

How to Select Dropdown in Selenium?

2. By creating Custom Locator and without iterating the List:

- In this method, you have to create custom Locator in Selenium, and then handle the drop down without iterating the List

#Method 2: By creating Custom Locator and without iterating the List

```
option1 = "Highest to lowest"
```

```
dropdown = driver.find_element(By.XPATH, value="//select/option[contains(text(),  
"+option1+"")]")
```

```
dropdown.click()
```

Why do we need selenium waits?

How to Select Dropdown in Selenium?

3. By using JavaScriptExecutor class:

- JavaScriptExecutor is an Interface that helps to execute JavaScript through Selenium WebDriver.
- You can use the `execute_script` method of JavaScriptExecutor interface to select an option from the drop down by using the value property of the element

#Method 3: By using JavaScriptExecutor class

```
option2 = driver.find_element(By.XPATH, value="//select")
```

```
driver.execute_script("arguments[0].value='highestprice'", option2)
```

Why do we need selenium waits?

How to Select Dropdown in Selenium?

4. By using send_keys

- You can use send_keys() method to select an option from a drop down by passing the element's value as sendKeys char sequence

#Method 4: By using send_keys method

```
driver.find_element(By.XPATH, value="//select").send_keys("lowestprice")
```


Why do we need selenium waits?

How to Select Dropdown in Selenium?

5. By using ActionChains Class

- Some drop-downs need to be hovered to display all the options present in it and then click on it.
- ActionChains class of Selenium is used for handling keyboard and mouse events

#Method 5: By using Actions Class

```
ele = driver.find_element(By.XPATH, value="//select")
action = ActionChains(driver)
action.move_to_element(ele).perform()
customOption = driver.find_element(By.XPATH, value="//option[text()='Lowest to highest']")
customOption.click()
```